Alumni Fundraising Project

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Summary:
In this project, I will conduct a general exploratory analysis on sample university alumni gift data to reveal trends in the gifts received and the types of donors who have given them. I will also use recency, frequency, and monetary (RFM) analysis to determine which donors my hypothetical Alumni Relations and Development department should prioritize with their outreach.

Data:
The dataset used for this project was generated by AI. I asked ChatGPT to provide code in Postgresql that would generate a relational database that fits the context of the Business Intelligence Analyst Associate position. To do this, I used keywords from the job listing as well as general information about Northwestern.

Here is part of my prompt: “The role is within the Alumni Relations & Development department of a T20 research university. The job summary specifically mentions that I'll be working with "fundraising and advancement" data.”

For more information on the relational database I used, please see the “AI Sample Database Tables - Snapshot” document.

*Please Note: While my dataset was generated with AI, **I certify that none of the code used in this project is AI generated**.

Analysis:
Below are the results of my SQL queries and insights. Please see my SQL script in the “alumni_fundraising_sample_script.sql” file.

–total all time gifts and the average gift amount

```
Total Gift Amount All Time|Average All Time Gift Amount|
--------------------------+----------------------------+
             $8,697,050.97|                     $434.85|
```

Insight: The university has received over $8.5 million in donations overall, and the average amount given by a donor is around $430.

–date range of gifts

Date of Earliest Donation	Date of Most Recent Donation
2016-03-18	2026-03-14

Insight: This dataset is reporting on around 10 years of donations.

–total gift amounts broken down by the type of gift

Type of Gift	Total Gift Amount
Recurring	$1,916,755.19
OneTime	$1,740,485.52
Planned	$1,727,332.51
Major	$1,695,362.08
Pledge	$1,617,115.67

Insight: Recurring gifts make up the largest portion of the total amount of gifts given.

–total gift amounts broken down by the major of the alumni donors

Major	Total Gift Amount
Engineering	$1,355,514.05
English	$1,010,816.43
Physics	$968,363.70
Statistics	$945,983.43
History	$921,881.23
Computer Science	$908,385.29
Chemistry	$889,195.59
Economics	$884,197.87
Biology	$780,541.66
Finance	$25,614.17
CS	$4,728.23
Mechanical Engineering	$1,172.68
Political Science	$656.64

Insight: Alumni who majored in Engineering have given the most in gifts, followed by English and Physics.

–total gift amounts broken down by the degree conferred

Degree	Total Gift Amount
BA	$2,184,345.42
MBA	$1,761,368.29
PhD	$1,664,080.03
BS	$1,617,070.45
MS	$1,470,186.78

–total gift amounts broken down by the age of the alumni donors

Age	Total Gift Amount	Number of Gifts
36	$584,814.56	651
53	$348,508.01	705
40	$319,345.25	602
25	$305,642.22	522
32	$280,428.59	621
52	$274,734.90	616
50	$271,198.81	649
48	$265,357.42	608
27	$252,912.53	593
38	$251,007.54	597
51	$247,105.30	632
43	$246,356.95	577
47	$244,386.44	525
55	$244,142.91	596
35	$241,730.12	620
37	$240,244.49	555
41	$239,687.49	611
22	$239,390.02	536
30	$231,256.51	611
28	$229,147.31	587
23	$225,795.17	604
39	$225,693.32	538
34	$223,072.99	538
29	$222,544.87	509
42	$219,274.43	556
44	$217,597.67	521
56	$210,577.83	523
54	$209,551.46	601
24	$208,802.62	559
49	$206,327.21	510
26	$202,475.89	507
33	$197,200.54	538
31	$192,285.41	519
45	$185,082.04	499
46	$167,757.98	461
61	$25,614.17	3

–extracting the min, max, and count of ages (from the previous results)

min	max	count
22	61	36

Insight: The age range of alumni donors spans from age 22 to age 61.

–top 10 ages based on the total amount of gifts given

Age	Total Gift Amount
25	$305,642.22
27	$252,912.53
32	$280,428.59
36	$584,814.56
38	$251,007.54
40	$319,345.25
48	$265,357.42
50	$271,198.81
52	$274,734.90
53	$348,508.01

Insight: The top 10 ages in terms of total amount of gifts given ranges from 25 to 53, with donors aged 36 having given the most in gifts.

–total amount of gifts since the beginning of the year (01/01/2026)

Total Gift Amount YTD
$1,282,221.81

Insight: We have received over $1.2 million in gifts since the beginning of the year.

–average number of months since last gift

Average Number of Months Since Last Gift
12

Insight: The average alumni donor has given their last gift about a year ago.

–total gift amounts broken down by the year given

Year	Total Gift Amount
2016	$393,853.65
2017	$511,978.95
2018	$439,889.97
2019	$453,984.38
2020	$491,254.05
2021	$579,517.52
2022	$623,958.43
2023	$864,841.90
2024	$1,104,811.12
2025	$1,942,742.46
2026	$1,290,218.54

Insight: There is generally an increasing trend in the amount of gifts received per year. We received our all-time high in gifts last year (in 2025).

–breakdown of total gifts received in 2025 and 2026 looking at month by month

Year	Month	Total Gift Amount
2025	1	$135,780.17
2025	2	$88,300.08
2025	3	$121,125.66
2025	4	$119,486.73
2025	5	$134,565.25
2025	6	$116,434.97
2025	7	$162,458.64
2025	8	$173,268.94
2025	9	$170,247.16
2025	10	$168,566.03
2025	11	$297,925.51
2025	12	$254,583.32
2026	1	$276,790.57
2026	2	$389,069.02
2026	3	$624,358.95

Insight: If you look at the first 3 months of 2025 and 2026, we have already received more in gifts this year than we did at this time last year. Assuming we stay on this trend, we can predict that we will earn more than $1.9 million in gifts this year.

Were the campaigns successful?
–total gift amounts broken down by campaign

Campaign	Total Gift Amount	Target Amount	Campaign End Date
Annual Giving FY26	$7,415,873.75	$2,000,000.00	2026-06-30
Campaign for Science Building	$30,000.00	$50,000,000.00	2027-12-31
Scholarship Drive 2025	$100.00	$500,000.00	2025-12-31
	$1,251,077.22		

Insight: The Annual Giving FY26 campaign is already successful a few months ahead of time. We haven't yet reached our goal for the Campaign for Science Building, but we still have until the end of next year. Unfortunately, we were unable to reach our goal of $500k for Scholarship Drive 2025.

Who are the top donors?
—top 5 donors by gift amount

constituent_id	first_name	last_name	Total Gift Amount
27	First17	Last17	$336,777.44
2683	First2653	Last2653	$100,656.15
1826	First1796	Last1796	$94,114.45
2156	First2126	Last2126	$55,962.72
202	First172	Last172	$52,623.23

Insight: Donor First17 Last17 has given the most in gifts with over $336,000k given over his lifetime.

–demographic information for those 5 donors

first_name	last_name	age	degree	grad_year	major	employer	job_title	Number of Donations	Total Gift Amount
First17	Last17	36	BA	1994	Engineering	University	Analyst	63	$336,777.44
First172	Last172	53	MBA	2002	Statistics	ConsultingCo	Manager	7	$52,623.23
First1796	Last1796	40	MBA	2012	English	ConsultingCo	Analyst	8	$94,114.45
First2126	Last2126	32	BA	1994	Physics	BioLabs	Manager	4	$55,962.72
First2653	Last2653	25	BA	2006	Engineering	University	Manager	7	$100,656.15

Insight: First 17 Last 17 is 36 years old, so they're likely skewing the data and making the total gift amount given by 36 year olds higher. Also, that person has given 63 gifts over the course of their life, which is much higher than the other top donors.

Other general findings: 2 out of the 5 donors majored in Engineering, and 3 out of the 5 donors are in manager positions.

Recency, Frequency, and Monetary (RFM) Analysis
This is the process of assigning three values to each donor based on their recency, frequency, and monetary scores, to help inform which donors my hypothetical Alumni Relations and Development department should prioritize.

Recency (Number of days since last gift given)
–percentiles for the recency analysis

tenth_percentile	thirtieth_percentile	median	seventieth_percentile	nintieth_percentile
3	28	102	299	1058

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they donated very recently (less than 3 days ago), and a value of 1 means they donated a long time ago (more than 2 years ago).

constituent_id	days_since_last_gift	recency_value
1489	894	2
4790	969	2
273	730	2
3936	1037	2
2574	161	3
951	254	3
4326	90	4
2614	918	2
2520	32	4
2466	60	4

Frequency (The number of gifts given)
–percentiles for the frequency analysis

tenth_percentile	thirtieth_percentile	median	seventieth_percentile	nintieth_percentile
2	3	4	5	7

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they have given a large number of gifts (more than 7), and a value of 1 means they have only given 1 gift (less than 2).

constituent_id	count_gifts	frequency_value
1489	5	4
4790	6	5
273	4	3
3936	2	1
2574	3	2
951	2	1
4326	7	5
2614	1	1
2520	6	5
2466	5	4

Monetary (The total dollar amount of gifts given)
–percentiles for the monetary analysis

tenth_percentile	thirtieth_percentile	median	seventieth_percentile	nintieth_percentile
367.72	739.25	1093.61	1600.35	4107.65

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they have given a larger dollar amount in gifts (more than ~$4100), and a value of 1 means they have given a smaller dollar amount in gifts (less than ~$360).

constituent_id	sum_gifts	monetary_value
1489	1934.77	5
4790	1994.47	5
273	974.38	3
3936	277.35	1
2574	637.08	2
951	1817.87	5
4326	2554.63	5
2614	357.13	1
2520	1963.97	5
2466	1246.44	4

–After combining all of these results into a new table that has four columns, the donor's ID and a column for each of their scores, we can filter for the donors who have 6's in all three categories. Here is the result of that query.

constituent_id	recency_value	frequency_value	monetary_value
4723	6	6	6
4254	6	6	6
32	6	6	6
2277	6	6	6
4784	6	6	6
3707	6	6	6
2072	6	6	6
1253	6	6	6
4923	6	6	6
490	6	6	6
1826	6	6	6
625	6	6	6
2158	6	6	6
327	6	6	6
1175	6	6	6
2442	6	6	6
1344	6	6	6

Insight: There are 17 donors who have given a large number of gifts, and high value of gifts, and have given a gift recently. Essentially, these are our most engaged and supportive donors.

–the demographic information of these 17 donors

constituent_id	first_name	last_name	age	degree	grad_year	major	employer	job_title
490	First460	Last460	22	BS	2009	History	RetailCo	Manager
625	First595	Last595	23	MBA	1990	Engineering	University	Manager
1253	First1223	Last1223	27	BA	2014	Computer Science	BioLabs	Researcher
3707	First3677	Last3677	31	BS	2012	Computer Science	FinanceGroup	Director
2277	First2247	Last2247	35	MS	2010	Physics	ManufacturingCo	Researcher
2442	First2412	Last2412	37	MS	2009	Computer Science	ManufacturingCo	Owner
4723	First4693	Last4693	39	PhD	2013	Economics	BioLabs	Manager
4923	First4893	Last4893	39	MS	1998	Statistics	FinanceGroup	Consultant
1826	First1796	Last1796	40	MBA	2012	English	ConsultingCo	Analyst
2072	First2042	Last2042	40	PhD	2004	Computer Science	University	Director
32	First2	Last2	45	BS	2004	History	FinanceGroup	Manager
2158	First2128	Last2128	46	PhD	1990	History	FinanceGroup	Consultant
1175	First1145	Last1145	48	PhD	2010	Statistics	TechCorp	Analyst
1344	First1314	Last1314	48	BA	2005	Engineering	RetailCo	Owner
4254	First4224	Last4224	49	MS	2013	Economics	University	Analyst
4784	First4754	Last4754	55	MBA	1994	Physics	TechCorp	Engineer
327	First297	Last297	56	BA	2012	English	ManufacturingCo	Owner

Insight: This group ranges in age from 22 to 56, with most of them being in their 30s and 40s. This could signal that alumni in their 30s and 40s are more likely to be engaged and supportive donors. Also, more than half of the donors in this group have some sort of leadership role, whether that be manager, director, or owner. Therefore, it's likely that targeting donors in those types of roles would be best.

Conclusion:

Overall, my hypothetical Alumni Relations and Development department is doing very well. We have been accepting gifts for 10 years and we've generally been receiving more in gifts each year. Although we weren't able to reach our goal for our campaign last year, we're crushing the Alumni Giving FY26 one for this year, and since we're projected to do better this year than last year, it's likely that we'll crush the Campaign for Science Building one, ending next year, too.

The RFM analysis revealed 17 donors who we should prioritize our outreach to. It also revealed trends in the types of donors that might fall into this group in the future. These types of donors are typically in their 30s and 40s, and have some sort of leadership role (e.g. manager, director, owner).

If you took a look at the schema diagram for this relational database, you noticed that there are additional tables that I didn't use in this project. Those tables simply didn't have enough data to be realistic, and I didn't want this report to be too long. If I were to build upon my work, I would expand the other tables, and look for trends related to who attended our events and then subsequently donated, which events garnered the most in gifts, which interactions/communications with donors were the most successful, and if there is a pattern between donors who engage/support us the most and where they live.